# Spatiotemporal Groundwater Level Prediction Using Machine Learning for Location-Aware Aquifer Monitoring

**Proposed Research Methodology: Karnataka, India**

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**Objective.** The study will combine incomplete groundwater records, environmental drivers, and spatial relationships to forecast future levels, reconstruct missing periods, and predict at locations without groundwater histories. The research gap will be addressed through an inductive spatiotemporal model, explicit missing-input uncertainty, and spatially blocked evaluation.

## A. Overall System Architecture

**Inputs** will include groundwater measurements and timestamps, well coordinates, meteorological histories, and verified static attributes. **Processing** will include quality control, monthly aggregation, spatial matching, missing-data reconstruction, temporal encoding, graph message passing, and uncertainty calibration. **Outputs** will be groundwater-depth estimates, nominal 90% prediction intervals, reconstructed observations, and spatial error summaries.

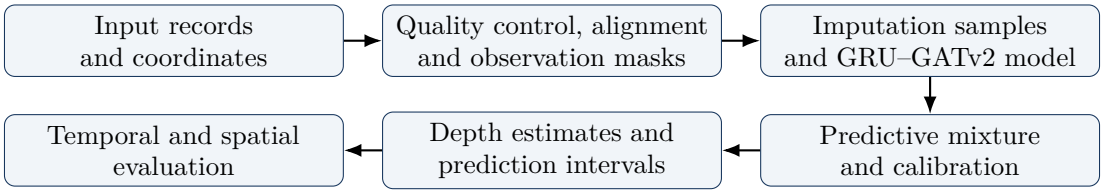



Figure 1: Proposed workflow. Learned preprocessing and model fitting will use training data only.

Three tasks will be evaluated separately: forecasting at monitored wells, retrospective gap reconstruction, and no-history prediction at spatial query locations. Forecasts will use only information available at the prediction date. Reconstruction may use observations before and after a historical gap. The primary horizon will be one month; direct three- and six-month forecasts will be secondary experiments where coverage permits.

## B. Dataset Description

**Source and study period.** The intended period will be January 2021–December 2025. Groundwater records will be obtained from Karnataka telemetry resources on the National Water Data Portal (NWDP), maintained by the National Water Informatics Centre. The portal identifies 2021–2025 groundwater resources and Karnataka rainfall listings [1]. Actual record coverage, cadence, coordinates, units, and measurement references will be audited before modelling.

**Target and sample unit.** The intended target will be groundwater depth in metres below ground level, after verifying the source definition. Smaller depth will indicate shallower groundwater; this quantity will not be interpreted directly as hydraulic head. Each modelling row will represent one canonical well and month. Source networks will be deduplicated only when physical well identity is verified.

**Number of samples.** The raw datasets have not yet been ingested, so empirical counts are pending. The data inventory will report unique wells, raw measured records, valid observed well–month targets, missingness, and usable windows in each partition. With $N$ wells and 60 calendar months, $60N$ is the possible grid size, not the measured sample count. A complete series with lookback $L$ and horizon $h$ will provide at most $60 - L - h + 1$ windows: 48 for $L = 12$ and $h = 1$. Imputed targets will not be counted as measured samples.

Table 1: Input groups and proposed features.

| Group | Variables | Role |
|---|---|---|
| Groundwater | Depth, timestamp, well ID, coordinates, observation mask | Target, historical state, and monitoring availability |
| Meteorology | Rainfall, temperature; other verified weather variables | Recharge and evaporative-demand indicators |
| Static attributes | Coordinates, elevation; verified basin, lithology, aquifer and well attributes | Spatial context and site differences |
| Optional drivers | Discharge, reservoir/tank information, satellite/reanalysis, land use, abstraction proxies | Feature-group comparisons, subject to source verification and overlap |

**Preprocessing will proceed as follows:**

1. Audit identifiers, coordinates, timestamps, units, duplicates, sensor errors, and measurement-reference changes. Flag suspicious values; exclude them only with a documented reason.
2. Convert timestamps consistently to Asia/Kolkata. Reduce telemetry to daily medians, then monthly medians. Require at least ten valid observation days per telemetry month; other targets will remain missing. Sparse seasonal measurements will not become measured monthly labels through interpolation.
3. Aggregate rainfall using verified incremental/cumulative definitions and retain monthly coverage. Map meteorological stations to well coordinates using up to five available neighbours and inverse-distance weighting with power two. Use geodesic distances; join station IDs directly only for verified shared installations.
4. Preserve original target-observation masks and separate input-hiding masks. Fit scaling, imputers, feature selection, and empirical thresholds within each training fold. Use global training-target scaling, since query wells have no history for per-well scaling.
5. Fill missing environmental inputs with their latest known value for at most one month, then a training-only calendar-month median, always retaining availability flags. Missing static values will use training medians or an unknown category.

The core training cohort will require at least 70% valid monthly targets during 2021–2023. Other training wells will support a separate sparse-data analysis. Test wells will be scored wherever genuine observations exist.

## C. Proposed Model / Algorithm

**Feature engineering.** Historical groundwater, input masks, rainfall and temperature sequences, trailing three- and six-month rainfall totals, and seasonal sine/cosine encodings will form the initial dynamic inputs. Coordinates and verified static attributes will supply spatial context. Lagged relationships and validation-based feature-group ablation will guide selection; weak instantaneous correlation alone will not justify removal.

**Temporal and spatial learning.** A GRU will encode a six- or twelve-month history, followed by two GATv2 layers and an MLP output head. Graph learning is motivated by multi-well studies in the submitted survey [2], while GATv2 supports query-dependent attention [3]. Let $y_{i,t}$ denote depth, $r_{i,t}$ the input mask, $\boldsymbol{x}_{i,t}$ the environmental vector, and $\boldsymbol{a}_i$ the static attributes:

$$\boldsymbol{z}_{i,t} = \mathrm{GRU}\left([\widetilde{y}_{i,\tau}, r_{i,\tau}, \boldsymbol{x}_{i,\tau}]_{\tau=t-L+1}^{t}\right), \tag{1}$$

$$\boldsymbol{g}_{i,t} = \mathrm{GATv2}^{(2)}\left([\boldsymbol{z}_{i,t}, \boldsymbol{a}_i], \mathcal{G}\right), \tag{2}$$

$$(\mu_{i,t,h}, v_{i,t,h}) = \mathrm{MLP}(\boldsymbol{g}_{i,t}, c_{t+h}). \tag{3}$$

Here $c_{t+h}$ is the known target-month encoding. Future observed environmental values will not enter forecasting. The variance output will use softplus plus $10^{-4}$ on the standardized scale.

Directed edges will connect each target to $k$ nearest eligible monitored wells, initially $k = 5$, with self-loops. Edge attributes will be $e_{ij} = \exp[-d_{ij}^2/(2\ell^2)]$, where $\ell$ is the median positive training-graph neighbour distance. They will be supplied explicitly to GATv2. The first layer will concatenate attention heads with ELU; the second will average heads. A 32-unit ReLU MLP will predict mean and variance. Edges will represent statistical proximity, not verified flow paths.

**Training objective.** A Gaussian output distribution will be fitted using observed targets only. For original observation mask $o_{i,t+h}$:

$$\mathcal{L}_h = \frac{\sum_{i,t} o_{i,t+h} \left[\log(2\pi v_{i,t,h}) + (y_{i,t+h} - \mu_{i,t,h})^2 / v_{i,t,h}\right]}{2 \sum_{i,t} o_{i,t+h}}. \tag{4}$$

Imputed targets will never receive an observed-target mask. The conditional Gaussian assumption will be checked using residual and interval diagnostics.

**Prediction without target-well history.** A query node will receive environmental history, static attributes, and an all-missing groundwater mask, with zero placeholders on the standardized scale. It will receive neighbour messages but will not provide messages to other nodes. No well-specific learned ID embedding will be used. During training, 20% of target wells per batch will act as pseudo-queries: their complete groundwater input histories and derived features will be hidden before imputation and graph construction, while observed future targets remain available for the loss. This will teach the model the inference condition. The task is stricter than excluding a well from training while providing its groundwater history at inference, as in Zhuang et al. [4].

**Missing-data uncertainty.** DeepMVI will be fitted on training records using artificial isolated and block masks [5]. Its conditional Gaussian sampling model will use residual variance estimated from an internal 10% masked-training subset. For forecasting, reconstruction inputs will be truncated at each origin; retrospective reconstruction may use surrounding observations. Entirely unmonitored query histories will not be imputed into groundwater features.

Twenty reconstructed histories will be sampled at inference, with measured values fixed. Their forecast distributions will form

$$p(y \mid \mathcal{I}_t) \approx \frac{1}{20} \sum_{b=1}^{20} \mathcal{N}(y; \mu^{(b)}, v^{(b)}). \tag{5}$$

The point estimate will be the mean of component means; base 90% limits will be the 0.05 and 0.95 mixture quantiles. Separate calibration sets will expand intervals using scores $s_j = \max(L_j - y_j, y_j - U_j, 0)$ and the $\lceil (n_{\text{cal}} + 1)0.90 \rceil$-th ordered score. Temporal and spatial dependence may violate conformal exchangeability assumptions [6]; empirical coverage will therefore be reported without an unconditional guarantee. Sampling will propagate the specified imputation uncertainty, rather than claim a complete epistemic-uncertainty decomposition.

## D. Training & Testing Strategy

**Spatial split.** Wells will be grouped into 25-km cells using EPSG:32643 coordinates and cell indices $(\lfloor x/25000 \rfloor, \lfloor y/25000 \rfloor)$. Occupied cells will be sorted and shuffled with seed 42. Whole cells will be assigned sequentially to approximately 20% test wells, then 16% spatial-validation wells, leaving approximately 64% training wells. Actual assignments will be recorded. Validation/test wells will enter only as query nodes, with all groundwater histories hidden. Geodesic distances will define graph edges and station matching.

**Validation and tuning.** Rolling-origin development checks will fit through December 2022 and validate January–June 2023, then fit through June 2023 and validate July–December 2023. Every learned transformation will be refitted inside its fold. Twelve configurations will be sampled using

Table 2: Planned chronological split, subject to verified coverage.

| Stage | Period | Use |
|---|---|---|
| Initial fitting | 2021–2023 | Training wells; preprocessing, imputer and model fitting |
| Validation/tuning | Jan.–Jun. 2024 | Monitored temporal validation and no-history spatial validation |
| Final refitting | 2021–Jun. 2024 | Training wells only; selected configuration |
| Calibration | Jul.–Dec. 2024 | Separate monitored and no-history interval calibration; no tuning |
| Final testing | 2025 | Future monitored forecasts and final unseen-well predictions |

GRU hidden sizes $\{32, 64\}$, lookbacks $\{6, 12\}$, neighbours $\{3, 5, 10\}$, heads $\{2, 4\}$, dropout $\{0.1, 0.3\}$, and learning rates $\{10^{-3}, 3 \times 10^{-4}\}$. Adam, weight decay $10^{-4}$, gradient clipping at 1, batches of four monthly graph snapshots, and at most 200 epochs will be used. Early stopping will use validation RMSE with patience 20; final refitting will use the selected epoch count.

Configurations will be ranked by the mean equal-well-weight RMSE across monitored and no-history validation tasks. Final fits will use seeds 42, 43, and 44 with separately calibrated intervals. Calibration and test observations will not select features or parameters. Test forecasts will begin after calibration, using newly available histories without updating model weights.

**Experiments and comparisons.**

1. Forecast monitored wells against persistence, seasonal naive, Ridge, Random Forest, XGBoost, and a GRU without graph learning.
2. Test no-history wells against environmental/static regressors and neighbour-persistence forecasts. IDW using target-date neighbour observations will be reported separately as contemporaneous spatial estimation.
3. Hide 10%, 20%, 30%, and 40% of observed values using isolated and one- to three-month block masks, with five realizations each. Score reconstruction only at hidden measured values; assess forecast degradation separately with scoring targets intact.
4. Remove graph learning, static attributes, pseudo-query training, or imputation sampling one at a time; test optional feature groups using the same partitions and comparable fitting budgets.

Source versions, quality-control rules, fold manifests, features, configurations, seeds, software versions, and predictions will be saved. Checks will confirm that query-well groundwater and post-origin observations never enter forecast inputs.

## E. Evaluation Metrics

The target is continuous, so regression metrics will be used:

$$\mathrm{MAE} = \frac{1}{n}\sum_j |y_j - \widehat{y}_j|, \qquad \mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_j (y_j - \widehat{y}_j)^2}, \tag{6}$$

$$\mathrm{NSE} = 1 - \frac{\sum_j (y_j - \widehat{y}_j)^2}{\sum_j (y_j - \overline{y})^2}. \tag{7}$$

MAE will measure typical error; RMSE will emphasize large errors. Both will be reported in metres after inverse scaling. NSE will compare predictions with the observed-mean benchmark. Standard SSE-based regression $R^2$ is identical on the same targets and will not be treated as independent evidence.

For intervals, prediction-interval coverage will be $\mathrm{PICP} = n^{-1}\sum_j \mathbf{1}\{L_j \le y_j \le U_j\}$, and mean width will be $\mathrm{MPIW} = n^{-1}\sum_j (U_j - L_j)$. Together these will assess reliability and informativeness relative to nominal 90% coverage. Base and calibrated intervals will be reported separately.

Each task/horizon will report target and well counts, pooled metrics, and equally weighted per-well metrics. Per-well NSE requires at least six targets and nonzero variance. Reconstruction scoring will use measured values. Three-seed mean and standard deviation, spatial errors, and distance-to-training-well summaries will assess stability and transfer.

## References

[1] National Water Informatics Centre, National Water Data Portal: Karnataka groundwater telemetry, Karnataka Ground Water Department telemetry, and Karnataka rainfall. Accessed 10 October 2026.

[2] M. L. Taccari et al., *Spatial-temporal graph neural networks for groundwater data*, Scientific Reports, 14, 24564, 2024. doi:10.1038/s41598-024-75385-2.

[3] S. Brody, U. Alon, and E. Yahav, *How Attentive are Graph Attention Networks?* arXiv:2105.14491, 2021. arXiv paper.

[4] C. Zhuang, L. Cui, and Y. Cui, *Enhancing groundwater level prediction with a hybrid deep learning model in Jinan City, China*, Scientific Reports, 15, 44535, 2025. doi:10.1038/s41598-025-28200-5.

[5] P. Bansal, P. Deshpande, and S. Sarawagi, *Missing Value Imputation on Multidimensional Time Series*, arXiv:2103.01600, 2021. arXiv paper.

[6] A. Auer et al., *A data-centric perspective on the information needed for hydrological uncertainty predictions*, Hydrology and Earth System Sciences, 28, 4099–4126, 2024. doi:10.5194/hess-28-4099-2024.